Notes: Aw, Roberts, and Xu (AER, 2011)

R&D Investment, Exporting, and Productivity Dynamics

Contents

1 Big picture

This paper studies the dynamic links among **exporting**, **R&D investment**, and **plant productivity**. Exporters are usually more productive than nonexporters. The core question is whether this is only selection, or whether exporting and R&D also change the future productivity path of firms.

The authors estimate a dynamic structural model using plant-level data from the Taiwanese electronics industry. In the model, plants decide whether to export and whether to conduct R&D. Both choices are costly, both can be persistent because of sunk costs, and both can affect future productivity.

Main finding

The paper finds that productivity evolves endogenously. Relative to plants that do neither activity, past exporting raises future productivity by about **1.96 percent**, past R&D raises future productivity by about **4.79 percent**, and doing both raises future productivity by about **5.56 percent**. R&D has the larger direct productivity effect, but exporting is cheaper and more common.

Economic message

The paper's central message is that trade liberalization can have long-run productivity effects beyond the immediate reallocation of production across firms. Larger export markets induce more exporting and more R&D; those choices improve future productivity; the productivity gains then further encourage exporting and R&D. This feedback loop is the main source of the gradual productivity response.

2 Incremental contribution of the paper

1. **Joint dynamic model of exporting and R&D.** Earlier work often studied exporting or R&D separately. This paper models both decisions together and allows both to affect future productivity.

2. **Endogenous productivity.** Productivity is not treated as a purely exogenous state variable. It evolves as a function of lagged productivity, lagged export participation, lagged R&D, and a stochastic innovation.

3. **Quantifies three channels.** The paper separates: (i) self-selection of high-productivity plants into exporting and R&D, (ii) direct effects of these choices on future productivity, and (iii) policy-induced changes in both activities when export-market size changes.

4. **Dynamic counterfactuals without a natural experiment.** The structural model is used to simulate the long-run response to an export-market expansion. This provides a way to study trade-liberalization effects even without a clean policy experiment in the data.

5. **Decomposes the role of R&D and exporting.** By comparing counterfactuals with and without learning-by-exporting or R&D effects, the paper shows that endogenous productivity nearly doubles the productivity gain from export-market expansion relative to a more restrictive model.

3 Research question and mechanism

3.1 Research question

How do exporting and R&D investment jointly determine plant productivity over time, and how does an expansion of export demand affect exporting, R&D, and productivity?

3.2 Core mechanism

Plants are heterogeneous in productivity, export demand shocks, size, and past experience. High-productivity plants have higher expected returns to exporting and R&D, so they self-select into both activities. Once they do, both activities can raise future productivity. This creates a reinforcing dynamic:

High productivity $\rightarrow$ export/R&D choice $\rightarrow$ future productivity gains $\rightarrow$ more export/R&D choice.

The model also includes fixed costs and sunk costs. A plant that already exports pays a fixed continuation cost, while a new exporter pays a sunk entry cost. The same structure applies to R&D. This generates persistence and hysteresis in both choices.

4 Structural model

4.1 Static revenue and profit

The plant has a short-run marginal cost function:

$$\ln c_{it} = \beta_0 + \beta_k \ln k_{it} + \beta_w \ln w_t - \omega_{it},$$

where k_{it} is capital, w_t is a vector of input prices, and ω_{it} is productivity. Higher productivity lowers marginal cost.

Demand is modeled separately for the domestic and export markets. Export revenue also includes a plant-specific export demand shock z_{it}. This matters because domestic and export revenues are not perfectly correlated; the model needs both productivity ω_{it} and the export demand shock z_{it} to explain revenue heterogeneity across markets.

4.2 Productivity transition

The key state equation is:

$$\omega_{it} = g(\omega_{i,t-1}, d_{i,t-1}, e_{i,t-1}) + \xi_{it},$$

with the empirical specification:

$$\omega_{it} = \alpha_0 + \alpha_1 \omega_{i,t-1} + \alpha_2 \omega_{i,t-1}^2 + \alpha_3 \omega_{i,t-1}^3 + \alpha_4 d_{i,t-1} + \alpha_5 e_{i,t-1} + \alpha_6 d_{i,t-1} e_{i,t-1} + \xi_{it}.$$

Here $d_{i,t-1}$ is lagged R&D participation and $e_{i,t-1}$ is lagged export participation. The parameters α_4, α_5, and α_6 measure how R&D, exporting, and their interaction affect future productivity.

4.3 Dynamic decisions

A plant chooses whether to export and whether to do R&D by comparing the marginal benefit of each activity with the relevant fixed or sunk cost.

For exporting, the marginal benefit is:

$$MBE_i(s_t \mid d_{i,t-1}) = \pi_{it}^X + V_{it}^E(s_t \mid d_{i,t-1}) - V_{it}^D(s_t \mid d_{i,t-1}).$$

For R&D, the marginal benefit is:

$$MBR_i(s_t \mid e_{it}) = E_t V_{i,t+1}(s_{i,t+1} \mid e_{it}, d_{it} = 1) - E_t V_{i,t+1}(s_{i,t+1} \mid e_{it}, d_{it} = 0).$$

A plant exports or conducts R&D if the relevant marginal benefit exceeds the cost draw. Costs are modeled as draws from exponential distributions.

5 Data

5.1 Setting

The empirical setting is the Taiwanese electronics industry, using plant-level data collected by Taiwan's Ministry of Economic Affairs for 2000 and 2002-2004. The survey was not conducted in 2001, so the model uses 2000 information to handle initial conditions and then studies 2002-2004 dynamic choices.

This industry is a useful laboratory because it is export-oriented, R&D-intensive, and central to Taiwan's manufacturing productivity growth. In 2000, electronics accounted for about 40 percent of Taiwanese manufacturing exports and more than 72 percent of manufacturing R&D expenditure.

5.2 Sample

The balanced panel contains **1,237 plants** that operate in all four sample years and report domestic sales, export sales, capital stocks, and R&D expenditure. The data are effectively plant-firm data: 91 percent of plants are owned by single-plant firms in the electronics industry.

Key descriptive facts:

- Roughly 60 percent of plants do not export in a given year.
- About 18.2 percent of plant-year observations report positive R&D expenditures.
- Exporting plants are much larger than nonexporters; export revenue is highly skewed.
- Many plants persist in their current status, especially plants that do neither activity, only export, or do both activities.

6 Estimation strategy

The model is estimated in two stages.

6.1 Stage 1: productivity and revenue parameters

The first stage estimates domestic revenue, cost, demand elasticities, and the productivity transition process. The domestic revenue equation is used to recover plant productivity. A key result is that both R&D and exporting have positive direct effects on future productivity, with R&D larger than exporting.

6.2 Stage 2: dynamic choice parameters

The second stage estimates the fixed and sunk cost distributions for exporting and R&D, as well as the export revenue shock process. The likelihood uses observed sequences of export participation, R&D participation, and export revenue.

The paper uses a Bayesian Markov chain Monte Carlo estimator to characterize the posterior distribution of the dynamic parameters.

7 Main empirical findings

7.1 Productivity effects

The estimated productivity process implies that, relative to plants doing neither activity:

- exporting raises future productivity by about **1.96 percent**;
- R&D raises future productivity by about **4.79 percent**;
- doing both raises future productivity by about **5.56 percent**.

R&D has the larger direct productivity effect. However, exporting is more common because export costs are lower than R&D costs.

7.2 Selection effects

The marginal benefits of both exporting and R&D rise strongly with current productivity. Therefore high-productivity plants are much more likely to self-select into both activities. This selection effect is the dominant driver of participation.

7.3 Costs and persistence

For both exporting and R&D, sunk entry costs are larger than fixed continuation costs. This explains persistence: plants already doing an activity are much more likely to continue than inactive plants are to enter.

The estimated costs of R&D are much larger than those of exporting. In the baseline model, the mean fixed cost of R&D is 67.606 million NT dollars and the mean sunk cost of R&D is 354.277 million NT dollars. The corresponding export costs are 11.074 and 50.753 million NT dollars.

7.4 Limited interdependence between R&D and exporting

A striking result is that the direct complementarity between the two activities is small. Prior R&D has little direct effect on the return to exporting, and past exporting has little direct effect on the return to R&D. The main connection between the two activities is indirect: they are both chosen by more productive plants and both affect future productivity.

7.5 Counterfactual export-market expansion

The counterfactual simulates a permanent increase in export market size. Under the full endogenous-productivity model, after 15 years:

- export participation rises by **10.2 percentage points**;
- R&D participation rises by **4.7 percentage points**;
- mean productivity rises by **5.3 percent**.

When productivity is exogenous, export participation still rises, but the long-run productivity gain disappears. This contrast shows why endogenous productivity is central to the paper's policy implications.

8 Tables from the paper

8.1 Descriptive evidence: sales and transition patterns

TABLE 1—DOMESTIC AND EXPORT SALES (*Millions of NT dollars*)

	Nonexporters	Exporters	
	Median domestic sales	Median domestic sales	Median export sales
2000	22.2	52.8	33.6
2002	17.0	36.4	32.5
2003	17.3	42.7	30.9
2004	17.8	38.6	30.7
	Average domestic sales	Average domestic sales	Average export sales
2000	69.5	390.0	586.0
2002	55.8	363.1	490.5
2003	57.2	385.9	576.3
2004	83.3	354.3	522.8

Table 1. Domestic and export sales.

TABLE 2—ANNUAL TRANSITION RATES FOR CONTINUING PLANTS

Status year t	Status year $t+1$			
	Neither	Only R&D	Only export	Both
All firms	0.563	0.036	0.255	0.146
Neither	0.871	0.014	0.110	0.005
Only R&D	0.372	0.336	0.058	0.233
Only export	0.213	0.010	0.708	0.070
Both	0.024	0.062	0.147	0.767

Table 2. Annual transition rates for continuing plants.

Read:

- Table 1 shows that exporters are much larger than nonexporters. Median domestic sales for exporting plants are roughly twice those of nonexporters, and average export sales are very large because export revenue is highly skewed.
- Table 2 shows strong persistence. Plants that do neither activity remain in that state with probability 0.871, plants that only export continue only exporting with probability 0.708, and plants that do both activities remain in both with probability 0.767.
- The transition matrix also shows that R&D-only plants are relatively rare and unstable; many stop R&D or move into both activities.

Economic message: Exporting and R&D are dynamic decisions with persistence, not one-shot static choices. The data require a model with sunk costs and joint transition dynamics.

8.2 First-stage productivity estimates and reduced-form participation

TABLE 3—DEMAND, COST, AND PRODUCTIVITY EVOLUTION
(*Parameter estimates—standard errors*)

Parameter	Discrete R&D	Continuous R&D
$1 + 1/\eta_D$	0.8432 (0.0195)*	0.8432 (0.0195)*
$1 + 1/\eta_X$	0.8361 (0.0164)*	0.8361 (0.0164)*
β_k	−0.0633 (0.0052)*	−0.0636 (0.0051)*
α_0	0.0879 (0.0198)*	0.0866 (0.0194)*
α_1	0.5925 (0.0519)*	0.5982 (0.0511)*
α_2	0.3791 (0.0915)*	0.3777 (0.0912)*
α_3	−0.1439 (0.0585)*	−0.1592 (0.0588)*
α_4	0.0479 (0.0099)*	0.0067 (0.0012)*
α_5	0.0196 (0.0046)*	0.0197 (0.0045)*
α_6	−0.0118 (0.0115)	−0.0022 (0.0014)
$SE(\xi_{it})$	0.1100	0.1098
Sample size	3,703	3,703

Table 3. Demand, cost, and productivity evolution.

TABLE 4—REDUCED FORM PARTICIPATION AND EXPORT REVENUE EQUATIONS

Dependent variable	Coeff on ω_{it}	Coeff on k_{it}	Coeff on e_{it-1}	Coeff on d_{it-1}	Other
Bivariate probit on exporting and R&D					
Exporting e_{it}	1.63 ($t = 10.3$)	0.064 ($t = 3.38$)	1.80 ($t = 32.1$)	0.186 ($t = 2.26$)	
R&D d_{it}	1.65 ($t = 7.12$)	0.205 ($t = 7.52$)	0.344 ($t = 4.38$)	1.86 ($t = 23.3$)	$\rho = 0.168$
Export revenue					
$\ln r_{it}^X$	6.45 ($t = 36.1$)	0.409 ($t = 20.3$)			
Export revenue with fixed effect (z_i)					
$\ln r_{it}^X$	5.55 ($t = 18.0$)	0.430 ($t = 4.16$)			$\mathrm{Var}(z) = 0.72$

Table 4. Reduced-form participation and export revenue equations.

Read:

- Table 3 shows that lagged R&D and lagged exporting both have positive and significant effects on future productivity. The R&D coefficient is larger than the export coefficient.
- The interaction term between R&D and exporting is negative but not statistically significant, suggesting limited complementarity in the productivity transition.
- Table 4 shows that current productivity is a strong predictor of both exporting and R&D. Productivity also strongly predicts export revenue.

Economic message: The key empirical facts are selection and learning. Productive plants are more likely to export and invest in R&D, and those activities also improve later productivity.

8.3 Dynamic parameters and model fit

TABLE 5—DYNAMIC PARAMETER ESTIMATES

Means and standard deviations of the posterior distribution					
Model 1			Model 2		
Parameter	Mean	Standard deviation	Parameter	Mean	Standard deviation
γ^I (R&D FC)	67.606	3.930	γ_1^I (size 1)	46.265	7.038
			γ_2^I (size 2)	66.596	3.423
γ^D (R&D SC)	354.277	31.377	γ_1^D (size 1)	381.908	66.521
			γ_2^D (size 2)	388.715	41.959
γ^F (Export FC)	11.074	0.389	γ_1^F (size 1)	5.733	0.295
			γ_2^F (size 2)	15.962	0.704
γ^S (Export SC)	50.753	3.483	γ_1^S (size 1)	51.852	6.046
			γ_2^S (size 2)	67.401	6.676
Φ_X (Export rev intercept)	3.813	0.063	Φ_X	3.873	0.063
ρ_Z (Export rev AR process)	0.773	0.014	ρ_Z	0.763	0.015
$\log \sigma_\mu$ (Export rev standard deviation)	−0.287	0.018	$\log \sigma_\mu$	−0.289	0.021

Table 5. Dynamic parameter estimates.

TABLE 6—R&D INVESTMENT RATES, EXPORT RATES, AND PRODUCTIVITY

	Year 2002	Year 2003	Year 2004
Export market participation rate			
Actual data	0.395	0.392	0.390
Predicted	0.370	0.371	0.371
R&D investment rate			
Actual data	0.177	0.170	0.169
Predicted	0.172	0.168	0.167
Average productivity			
Actual data	0.436	0.444	0.436
Predicted	0.449	0.441	0.432

Table 6. R&D rates, export rates, and productivity.

Read:

- Table 5 estimates fixed and sunk costs. For both R&D and exporting, sunk costs exceed fixed costs, consistent with persistence in participation.
- R&D costs are much higher than export costs. This helps explain why exporting is more common even though R&D has a larger productivity effect.
- Table 6 shows that the simulated model matches the observed export participation rates, R&D investment rates, and average productivity reasonably well for 2002-2004.

Economic message: The structural model captures both behavioral decisions and productivity evolution well enough to support the later counterfactual simulations.

8.4 Predicted transitions and marginal benefit of exporting

TABLE 7—PREDICTED TRANSITION RATES FOR CONTINUING PLANTS—MODEL 2

Status year t		Status year $t+1$			
		Neither	Only R&D	Only export	Both
Neither	Predicted	0.866	0.019	0.110	0.008
	Actual	0.871	0.014	0.110	0.005
Only R&D	Predicted	0.476	0.214	0.116	0.193
	Actual	0.372	0.336	0.058	0.233
Only export	Predicted	0.292	0.010	0.622	0.077
	Actual	0.213	0.010	0.708	0.070
Both	Predicted	0.049	0.028	0.138	0.784
	Actual	0.024	0.062	0.147	0.767

Table 7. Predicted transition rates for continuing plants.

TABLE 8—MARGINAL BENEFIT OF EXPORTING (*Millions of NT dollars*)

	V_t^E		V_t^D		$MBE = \pi_t^X + V_t^E - V_t^D$	
ω_t	$d_{t-1}=1$	$d_{t-1}=0$	$d_{t-1}=1$	$d_{t-1}=0$	$d_{t-1}=1$	$d_{t-1}=0$
−0.19	132.5	132.4	130.9	130.7	2.1	2.1
−0.02	138.9	138.5	136.3	135.9	3.7	3.8
0.15	151.8	150.9	147.3	146.3	7.1	7.2
0.32	179.4	176.3	170.9	167.4	14.7	15.2
0.49	245.3	235.6	228.9	217.7	31.3	32.9
0.67	392.6	365.3	362.9	331.9	65.3	69.1
0.84	714.0	655.9	667.0	599.1	132.3	142.1
1.01	1,206.3	1,117.4	1,143.7	1,041.5	266.8	280.1
1.18	1,911.3	1,790.0	1,834.0	1,695.3	565.7	583.2
1.35	2,689.1	2,568.8	2,610.8	2,471.7	1,246.9	1,265.7

Table 8. Marginal benefit of exporting.

Read:

- Table 7 shows that the model reproduces the main transition patterns, especially for the large groups of plants that do neither activity or only export.
- Table 8 shows that the marginal benefit of exporting rises sharply with productivity. At high productivity, the value of exporting is much larger.

- Prior R&D status changes the marginal benefit of exporting only slightly, reinforcing the conclusion that direct cross-activity effects are modest.

Economic message: Current productivity, not past R&D status, is the central state variable driving export participation.

8.5 Export policy functions and realized costs

TABLE 9—EXPORT POLICY FUNCTIONS: $\Pr(e_t = 1 \mid e_{t-1}, d_{t-1})$

ω_t	$e_{t-1} = 1$, $d_{t-1} = 1$	$e_{t-1} = 0$, $d_{t-1} = 1$	$e_{t-1} = 1$, $d_{t-1} = 0$	$e_{t-1} = 0$, $d_{t-1} = 0$
−0.19	0.156	0.032	0.157	0.032
−0.02	0.246	0.055	0.248	0.056
0.15	0.365	0.098	0.369	0.099
0.32	0.518	0.178	0.523	0.181
0.49	0.710	0.309	0.718	0.317
0.67	0.895	0.485	0.907	0.503
0.84	0.969	0.653	0.980	0.693
1.01	0.988	0.749	0.994	0.785
1.18	0.997	0.835	0.999	0.873
1.35	0.997	0.874	0.999	0.901

Table 9. Export policy functions.

TABLE 10—COSTS OF EXPORTING AND R&D (*Millions of NT dollars*)

	Mean export costs among exporters[a]		Mean R&D costs among investors[b]	
ω_t	Fixed cost	Sunk cost	Fixed cost	Sunk cost
−0.19	0.97	1.04	1.32	1.34
−0.02	1.61	1.80	2.06	2.12
0.15	2.64	3.33	3.41	3.59
0.32	4.18	6.50	6.55	7.28
0.49	6.21	12.32	12.83	15.64
0.67	8.50	21.06	24.01	33.74
0.84	9.86	30.72	36.28	60.78
1.01	10.35	37.17	43.49	86.61
1.18	10.67	43.54	49.35	113.93
1.35	10.70	47.02	48.26	114.87

[a] For plants with $d_{t-1} = 1$.
[b] For plants with $e_t = 1$.

Table 10. Costs of exporting and R&D.

Read:

- Table 9 translates marginal benefits into export probabilities. Export probabilities increase with productivity and are much higher for prior exporters than for prior nonexporters.
- At productivity $\omega = 0.49$ and prior R&D equal to one, the probability of continuing to export is 0.710, while the probability of entering from nonexport status is 0.309.
- Table 10 shows that mean costs rise with productivity among plants that choose the activity, because high-productivity plants can profitably bear higher costs.

Economic message: Sunk entry costs create a large gap between continuation and entry probabilities. High-productivity plants can overcome higher costs, so selection into exporting and R&D is strong.

8.6 Marginal benefit and policy function for R&D

TABLE 11—MARGINAL BENEFIT OF R&D INVESTMENT (*Millions of NT dollars*)

	$EV_{t+1}(d_t = 1, e_t)$		$EV_{t+1}(d_t = 0, e_t)$		MBR	
ω_t	$e_t = 1$	$e_t = 0$	$e_t = 1$	$e_t = 0$	$e_t = 1$	$e_t = 0$
−0.19	142.2	140.7	139.3	137.6	2.8	3.2
−0.02	150.2	147.9	145.8	142.9	4.5	5.0
0.15	166.2	162.1	158.6	153.7	7.6	8.4
0.32	200.3	192.1	184.6	175.1	15.7	17.0
0.49	278.6	262.5	244.3	224.6	34.3	37.9
0.67	447.3	417.5	371.2	333.0	76.1	84.5
0.84	797.2	749.5	654.2	583.3	143.0	166.3
1.01	1,320.2	1,255.6	1,105.7	1,004.1	214.5	251.5
1.18	2,065.7	1,985.2	1,773.5	1,637.5	292.2	347.7
1.35	2,883.8	2,802.5	2,575.2	2,425.5	308.6	377.0

Table 11. Marginal benefit of R&D investment.

TABLE 12—R&D POLICY FUNCTIONS: $\Pr(d_t = 1 \mid e_t, d_{t-1})$

ω_t	$e_t = 1$, $d_{t-1} = 1$	$e_t = 1$, $d_{t-1} = 0$	$e_t = 0$, $d_{t-1} = 1$	$e_t = 0$, $d_{t-1} = 0$
−0.19	0.041	0.007	0.046	0.008
−0.02	0.063	0.011	0.070	0.012
0.15	0.101	0.018	0.109	0.020
0.32	0.182	0.036	0.192	0.039
0.49	0.329	0.075	0.347	0.082
0.67	0.566	0.155	0.590	0.169
0.84	0.777	0.265	0.812	0.296
1.01	0.872	0.360	0.899	0.399
1.18	0.940	0.455	0.959	0.504
1.35	0.928	0.451	0.951	0.505

Table 12. R&D policy functions.

Read:

- Table 11 shows that the marginal benefit of R&D increases sharply with productivity. The benefit is much larger for high-productivity plants.
- Exporting status changes the return to R&D only modestly. In the estimated model, the return to R&D is slightly higher for nonexporters, but the magnitudes are small.
- Table 12 shows that the probability of doing R&D rises with productivity and is much higher for plants that already conducted R&D in the previous period.

Economic message: Like exporting, R&D is mostly driven by productivity and persistence. Direct complementarity between exporting and R&D is not the central force.

8.7 Counterfactual export-market expansion

TABLE 13—PLANT RESPONSE TO EXOGENOUS INCREASE IN EXPORT MARKET SIZE

	Year			
	2	5	10	15
Endogenous productivity: $\omega_{it} = g(\omega_{it-1}, d_{it-1}, e_{it-1}) + \xi_{it}$				
Change in proportion of exporters	5.2	9.0	10.0	10.2
Change in proportion of R&D performers	2.5	3.5	4.1	4.7
Percentage change in mean productivity	0.5	1.5	3.7	5.3
Endogenous productivity: $\omega_{it} = g(\omega_{it-1}, d_{it-1}) + \xi_{it}$				
Change in proportion of exporters	2.0	3.9	4.0	4.4
Change in proportion of R&D performers	1.8	2.6	3.5	4.0
Percentage change in mean productivity	0.0	0.8	1.9	2.9
Endogenous productivity: $\omega_{it} = g(\omega_{it-1}, e_{it-1}) + \xi_{it}$				
Change in proportion of exporters	3.4	5.9	7.2	7.6
Percentage change in mean productivity	0.0	0.0	1.2	1.8
Exogenous productivity: $\omega_{it} = g(\omega_{it-1}) + \xi_{it}$				
Change in proportion of exporters	4.6	5.7	5.8	5.5

Table 13. Plant response to an exogenous increase in export market size.

Read:

- In the full endogenous-productivity model, export participation rises by 10.2 percentage points and R&D participation rises by 4.7 percentage points after 15 years.
- Mean productivity rises by 5.3 percent after 15 years in the full model.
- If productivity is exogenous, export participation rises by only 5.5 percentage points and mean productivity does not rise.
- If exporting has no direct productivity effect, the 15-year productivity gain falls to 2.9 percent. If R&D has no direct productivity effect, the gain falls to 1.8 percent.

Economic message: The long-run gains from export-market expansion depend heavily on endogenous productivity. R&D and learning-by-exporting create gradual productivity growth after the initial increase in export demand.

9 How to interpret the paper

9.1 What the paper supports

The evidence supports a model in which export participation and R&D are dynamic, costly, persistent choices. More productive plants self-select into these choices, and the choices themselves improve future productivity. Trade liberalization therefore affects productivity not only through reallocation toward productive exporters, but also through within-plant productivity improvements induced by export-market incentives.

9.2 What the paper does not claim

The paper does not claim that exporting and R&D are strong direct complements. In fact, the estimated interaction between lagged exporting and lagged R&D is small and statistically weak. The main connection is through productivity: both activities are more attractive to productive plants, and both help move the future productivity state.

9.3 Why the dynamic model matters

A static model would capture selection into exporting and R&D but miss the feedback loop. The dynamic model can trace how a policy shock changes choices today, how those choices change productivity tomorrow, and how improved productivity changes choices again in later years.

10 Bottom line

Aw, Roberts, and Xu show that exporting and R&D are jointly determined dynamic investments that shape productivity over time. In Taiwanese electronics, R&D has the larger direct productivity effect, exporting is cheaper and more prevalent, and high-productivity plants self-select into both. A larger export market increases exporting, induces more R&D, and generates gradual within-plant productivity growth. The paper's key contribution is to quantify this endogenous productivity channel in a structural framework.